Syed Salauddin Mohammad **Tariq**

PhD Candidate

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Research Interests

My research interests span the broad spectrum of computer systems, distributed systems, cloud computing, software engineering, and artificial intelligence (AI), with a focus on performance observability and optimization. I am passionate about advancing evidence-grounded, AI-assisted approaches for understanding and optimizing increasingly complex software and cloud systems.

Education

Ph.D. in Computer and Information Science — *Jan 2022 – Present*
University of Michigan Dearborn, Rackham Graduate School — *Dearborn, MI, USA*

- **Advisor:** Dr. Probir Roy
- **Dissertation:** Cross-Layer Performance Observability and Optimization in Cloud Systems (proposal exam passed Mar 2026)
- **Dissertation Committee:** Dr. Probir Roy (Chair), Dr. Zheng Song, Dr. Jinhua Guo, Dr. Junaid Farooq

Master of Science in Computer and Information Science — *Jan 2022 – Apr 2024*
University of Michigan Dearborn — *Dearborn, MI, USA*

- Graduated with High Distinction

Bachelor of Science in Computer Science and Engineering — *Jan 2011 – May 2015*
Patuakhali Science and Technology University — *Patuakhali, Bangladesh*

Publications

Conference Papers

2026 — Cold-Start Anti-Patterns and Refactorings in Serverless Systems: An Empirical Study,
***Syed Salauddin Mohammad Tariq**, Foyzul Hassan, Amiangshu Bosu, Probir Roy*;
IEEE International Conference on Software Analysis, Evolution, and Reengineering; Acceptance rate: **23%**; Rank: **A**; Status: **Published**; DOI: 10.1109/SANER67736.2026.00030
SANER '26 *Limassol, Cyprus*

2026 — LLM-Guided Mining of Performance-Related Commits at Scale,
*MAK Azad, **Syed Salauddin Mohammad Tariq**, Foyzul Hassan, Diego E. Costa, Probir Roy*;
International Conference on Evaluation and Assessment in Software Engineering;
Acceptance rate: **27%**; Rank: **A**; Status: **Accepted**
EASE '26 *Glasgow, SCT, UK*

2025 — Efficient Serverless Cold Start: Reducing Library Loading Overhead by Profile-guided Optimization,
***Syed Salauddin Mohammad Tariq**, Ali Al Zein, S. Vaidya, A. Khanolkar, Zheng Song, Probir Roy*;
IEEE International Conference on Distributed Computing Systems;
Acceptance rate: **19.7%**; Rank: **A**; Status: **Published**; DOI: 10.1109/ICDCS63083.2025.00037
ICDCS '25 *Glasgow, SCT, UK*

Peer-Reviewed Journal Articles

2023 — MicroProf: Code-level Attribution of Unnecessary Data Transfer in Microservice Applications,
***Syed Salauddin Mohammad Tariq**, Lance Menard, Pengfei Su, Probir Roy*;
ACM Transactions on Architecture and Code Optimization (**ACM TACO**) Journal, Vol. 20, No. 4, Article 50; Impact factor: **1.8**; Rank: **Q2**; Status: **Published**; DOI: 10.1145/3622787
TACO '23

Manuscripts in Preparation

2026 — PerfFinder: Navigating Configuration Search with Knob Influence Graphs,
***Syed Salauddin Mohammad Tariq**, Zheng Song, Probir Roy*;
Status: **In Preparation**; *to be submitted to The ACM Web Conference (**WWW 2027**)
*WWW '27** *Dublin, Ireland*

Poster

2024 — PcMINER: Mining Performance Related Commits at Scale,
*MAK Azad, M. Alexender, M. Alexender, **Syed Salauddin Mohammad Tariq**, F. Hassan, Probir Roy*
SC (formerly Supercomputing), the International Conference for High Performance Computing, Networking, Storage and Analysis; Status: **Published**;
sc24.supercomputing.org/proceedings/poster/poster_pages/post278.html
SC '24 *Atlanta, GA, USA*

Earlier Conference Papers

2017 — Cognitive complexity: A model for distributing equivalent programming problems,
*Samrat Kumar Dey, **SSM Tariq**, Md Shariful Islam, Golam Md. Muradul Bashir*;
International Conference on Electrical, Computer and Communication Engineering (**ECCE 2017**);
DOI: 10.1109/ECACE.2017.7913018
ECCE '17 *Cox's Bazar, BD*

2014 — Complexity measurement: A new approach to ensure equal distribution of programming problems for evaluation,
*GMM Bashir, Samrat Kumar Dey, **SSM Tariq**, Md Shariful Islam*;
International Conference on Electrical and Computer Engineering (**ICECE 2014**); DOI: 10.1109/ICECE.2014.7026848
ICECE '14 *Dhaka, BD*

Teaching Record

University of Michigan Dearborn

Fall 2026 — **Lab Instructor**, CIS-200L – Computer Science II Lab (27 students)
Winter 2026 — **Teaching Assistant**, CIS-450 – Operating Systems (54 students)
Fall 2025 — **Teaching Assistant**, CIS-450 – Operating Systems (42 students)
Winter 2025 — **Teaching Assistant**, CIS-450 – Operating Systems (62 students)
Fall 2024 — **Lab Instructor**, CIS-200L – Computer Science II Lab (27 students)
Summer 2024 — **Instructor of Record**, CIS-200 – Computer Science II (15 students)
Winter 2024 — **Teaching Assistant**, CIS-578 – Advanced Operating Systems (41 students)
Fall 2023 — **Teaching Assistant**, CIS-450 – Operating Systems (45 students)
Summer 2023 — **Lab Instructor**, Tools for Software Engineering, ESIEA Exchange Program (27 students)
Winter 2023 — **Teaching Assistant**, CIS-581 – Computational Learning (61 students)
Fall 2022 — **Teaching Assistant**, CIS-450 – Operating Systems (52 students)

Experience

Graduate Student Instructor — *Aug 2022 – Present*
University of Michigan Dearborn — *Dearborn, MI, USA*

- **Instructor of Record, CIS-200:** Developed and delivered lectures, assignments, and exams for a programming and software-engineering course, strengthening students' implementation, debugging, and problem-solving skills; overall instructor rating of **4.75/5** in student evaluations (department mean 4.51).
- **Lab Instructor, CIS-200L:** Led hands-on laboratory sessions, guiding programming exercises, debugging practice, and project implementation; sole instructor of one lab section each term, and designed all lab materials and programs.
- **Lab Instructor, ESIEA Exchange Program:** Taught engineering students from École Supérieure d'Informatique, Électronique, Automatique (ESIEA), **France**, introducing software engineering tools, workflows, and professional best practices.
- **Teaching Assistant, CIS-450, CIS-578, CIS-581:** Supported these courses through office hours, student support, and grading, and proctored exams for CIS-450 in all five terms.
- **Online/Hybrid Teaching:** Taught the online asynchronous section of CIS-200 and the students who joined CIS-200L online, and supported the online asynchronous sections of CIS-450 and CIS-578: recorded sessions, held online office hours, graded online submissions, and managed Canvas course sites, using Zoom, Mediasite, zyBooks, Copyleaks, and Turnitin.

Graduate Student Research Assistant — *Jan 2022 – Present*
University of Michigan Dearborn — *Dearborn, MI, USA*

- Developed **MicroProf**, a dynamic program-analysis tool that identifies unnecessary data transfer in Java microservices; its guided API optimizations reduced REST response fields by up to **87.5%** and improved service performance by up to **4.59x**.
- Developed **SlimStart**, a profile-guided optimizer for workload-dependent library initialization in Python serverless apps; achieved up to **2.30x** faster initialization, a **2.26x** improvement in end-to-end latency, and **1.51x** lower memory usage.
- Led SlimStart through the NSF I-Corps program as Entrepreneurial Lead, interviewing **20+** industry stakeholders.
- Investigated developer-visible serverless cold-start anti-patterns through an empirical study of **81** adjudicated issues across **35** repositories; co-developed **SCABENCH**, a reproducible benchmark of cold-start scenarios.
- Co-authored **PerfMiner**, an LLM-guided framework for mining performance-related software changes; the study analyzed approximately **170 million** commits across **323,700** repositories and produced an audited corpus of **335,103** commits.
- Conducting research under the supervision of Dr. Probir Roy.

Senior Software Engineer (Software Engineer until Oct 2018) — *Oct 2016 – Dec 2021*
BJIT Limited — *Dhaka, Bangladesh*

- Engineered and maintained Spring Boot REST APIs and Mule ESB integrations for enterprise clients, with CI/CD, code quality, testing, and monitoring workflows; received a Certificate of Appreciation (2019) from Rakuten Inc. for contributing greatly to the Core Architecture Section.
- Improved fault-localization workflows by implementing distributed tracing and Kibana visualizations; designed and validated APIs through integration, regression, load, system, and stress testing.

- Coordinated agile delivery through daily huddles, sprint retrospectives, offshore client communication, requirement analysis, estimation, negotiation, and sprint progress reporting.
- Designed programming problems for initial screening, administered in-person hiring exams, and conducted oral interviews for entry-level recruitment, including for the Bangladesh-Japan ICT Engineers' Training Program (B-JET), 2018–2021.

Software Engineer *May 2015 – Oct 2016*
BRAC IT Services Ltd. *Dhaka, Bangladesh*

- Implemented an automated CRM module for sbiCloud ERP, that improved customer response time by **70%**.
- Supported ERP deployment across **17** countries, contributing to a **25%** reduction in logistics costs for BRAC, the world's largest NGO.

Presentations

- **Conference Talk:** "*Cold-Start Anti-Patterns and Refactorings in Serverless Systems: An Empirical Study*," SANER 2026 *Limassol, Cyprus*
- **Conference Talk:** "*MicroProf: Code-level Attribution of Unnecessary Data Transfer in Microservice Applications*," HiPEAC 2024 *Munich, Germany*
- **Guest Lecture:** "*Memory Virtualization*," CIS-450 – Operating Systems (Winter 2026), University of Michigan Dearborn *Dearborn, MI, USA*
- **Poster:** "*ColdSpy: Reducing Library Loading Overhead by Profile-guided Optimization*," University of Michigan Dearborn CECS Doctoral Poster Competition 2025 *Dearborn, MI, USA*
- **Poster:** "*MicroProf: Code-level Attribution of Unnecessary Data Transfer in Microservice Applications*," HiPEAC 2024 *Munich, Germany*
- **Conference Talk:** "*Complexity measurement: A new approach to ensure equal distribution of programming problems for evaluation*," ICECE 2014 *Dhaka, BD*

Awards and Grants

Honors and Awards

2025	**Best Poster**, University of Michigan Dearborn CECS Doctoral Poster Competition, for the poster "*ColdSpy: Reducing Library Loading Overhead by Profile-guided Optimization*"	*Dearborn, MI, USA*
2022	**Honorable Mention (Originality)**, 2nd Student Design Competition on Networked Computing on the Edge, CPS-IoT Week 2022, for the team project UAV-Assisted Event Detection for Ground-Vehicles	*Milan, Italy*
2022	**Certificate of Appreciation**, University of Michigan Dearborn College of Engineering and Computer Science, for outstanding contributions to teaching, research, scholarship, and service	*Dearborn, MI, USA*

Travel Grants

2025	NSF Student Support Grant to attend ICDCS 2025 in Glasgow, Scotland, UK	*Dearborn, MI, USA*
2025	EXP+ Student Independent Research Grant 2024–25 to attend ICDCS 2025 in Glasgow, SCT, UK	*Dearborn, MI, USA*
2024	CECS Doctoral Student Conference Travel Grant to attend HiPEAC 2024 in Munich, Germany	*Dearborn, MI, USA*
2024	EXP+ Student Conference Presentation Grant 2023–24 to attend HiPEAC 2024 in Munich, Germany	*Dearborn, MI, USA*
2022	Bangladesh Sweden Trust Fund (BSTF) Student Grant, to begin Ph.D. at UM-Dearborn	*Dhaka, BD*

Service

Research Community Service

2026	**Artifact Evaluation Committee Member**, The Journal of Systems Research (**JSys**)	
2025	**Artifact Evaluation Committee Member**, USENIX Symposium on Operating Systems Design and Implementation (**OSDI**)	*Boston, MA, USA*
2024	**Artifact Evaluation Committee Member**, USENIX Symposium on Operating Systems Design and Implementation (**OSDI**)	*Santa Clara, CA, USA*

Journal Reviewing

2026	**Sub-reviewer**, IEEE Transactions on Cloud Computing (**TCC**)
2024–25	**Sub-reviewer**, ACM Transactions on Architecture and Code Optimization (**TACO**)

Academic Leadership and Outreach

2025–	**President**, ACM Student Chapter at University of Michigan Dearborn Revitalized the chapter Organized "*Research Like a Pro: A Hands-on Workshop for Students*" with **61** participants Launched the "*ACM Newsletter Service for the University of Michigan Dearborn*"	*Dearborn, MI, USA*
2022–	**Member**, Doctoral Student Association (**DSA**), University of Michigan Dearborn	*Dearborn, MI, USA*
2018–19	**Member**, **EducationUSA**, an education network supported by the U.S. Department of State	*Dhaka, BD*

Professional Memberships

2023– **Member**, Association for Computing Machinery (**ACM**) *Dearborn, MI, USA*
2023– **Member**, Institute of Electrical and Electronics Engineers (**IEEE**) *Dearborn, MI, USA*

Technical and Research Skills

Research Methods	Dynamic program analysis, performance profiling and attribution, distributed tracing, empirical software engineering, benchmark design, artifact evaluation, user-study evaluation, mining software repositories
AI and Software Analytics	LLM-assisted issue and commit analysis, transformer-based classification, dataset curation and validation
Cloud and Systems	Serverless computing, microservices, AWS Lambda, S3, SQS, IAM, EC2, Docker, Kubernetes, CloudLab, Linux, OpenTelemetry, Jaeger, Prometheus, Grafana, Kibana
Programming and Frameworks	Java, Python, Groovy, SQL, Spring Boot, Grails, Mule ESB, Hibernate, REST APIs
Data and Storage	PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, JDBC, SQLite
Machine Learning Tools	PyTorch, Scikit-learn, Keras, MATLAB, Pandas, NumPy, Matplotlib, Seaborn
Development and Testing	Git, Jenkins, SonarQube, Maven, JUnit, Postman, JMeter, Swagger, JIRA, Confluence

Projects

MicroProf: Diagnose Inefficient Data Transfer among Java Microservices *Dearborn, MI, USA*

- Uses Jaeger tracing, remote-object sampling, and hardware debug registers to attribute unused object fields to code.
- Associated publication: ACM TACO, 2023; artifact: figshare.com/s/5156839e442f03d97747.

SlimStart: Profile-Guided Optimizer for Serverless Cold Starts *Dearborn, MI, USA*

- Profiles library use by call path, lazy-loads rarely used libraries, and re-optimizes in CI/CD as workloads change.
- Evaluated on three benchmark suites and four real-world serverless applications.
- Associated publication: IEEE ICDCS, 2025.

SCABENCH: Cold-Start Benchmark *Dearborn, MI, USA*

- Isolates initialization inefficiencies in **18** executable Python scenarios, each with ground truth for evaluating tools.
- Associated publication: IEEE SANER, 2026; artifact: doi.org/10.5281/zenodo.17375894.